

Trainer Debrief — Lab 2

Portfolio Selection, Strategy Alignment and the Black Belt Role

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Align a project portfolio to business strategy and define the Black Belt's leadership role (A1, K1).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-02-portfolio-and-belt-roles.xlsx

Two projects ('Warehouse 5S rollout', 'Canteen menu survey') map to NO strategic pillar and should be stopped or handed to local management — they consume belt capacity for no strategic return. The portfolio is also unbalanced: 4 of 8 projects sit under Quality and none under Growth. On the role matrix, the Black Belt owns cross-functional DMAIC leadership, advanced statistics AND Green Belt mentoring — mentoring is the duty most often forgotten, and it is what distinguishes a Black Belt from a very good Green Belt.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every candidate project traces to a named business objective, your belt matrix distinguishes Black from Green Belt accountability, and your capstone has a defined five-tollgate governance plan.

Where this sits in the capstone

A Green Belt is handed a project; a Black Belt decides which projects exist. Practise the portfolio view — trace every candidate up to a business objective, balance the portfolio across DMAIC maturity and risk, and articulate what you as Black Belt own that a Green Belt does not.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27
Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30

If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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